Bidirectional Entity Coupling Alignment for Fine-Grained Remote Sensing Image–Text Retrieval

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***Abstract*—Due to the homogeneous backgrounds and cluttered layouts of remote sensing (RS) imagery, fine-grained local entities are indispensable for preventing cross-modal semantic mismatch. However, existing parameter-efficient fine-tuning (PEFT) methods based on contrastive language-image pretraining (CLIP) encapsulate scenes into holistic global representations, diluting these critical local cues. To address this, we propose the bidirectional entity-level alignment network (BEAN), a unified fine-grained entity alignment paradigm. First, to avoid the semantic ambiguity of conventional extraction mechanisms, we introduce a symmetric visual and textual entity construction paradigm comprising a visual entity bank constructor (VEBC) and a noun-to-token index constructor (NTIC). Specifically, VEBC constructs a visual entity bank via layerwise extraction, while NTIC performs index-guided extraction of textual entity embeddings, jointly delivering semantically discriminative cross-modal entity pairs. Second, to counteract the submersion of local entities within global embeddings, we design the cross-modal entity-bidirectional adapter (CEBA). It drives bidirectional cross-attention between the global feature stream and localized entity tokens. This mutually enriches isolated entities with holistic scene context, while simultaneously reinjecting fine-grained entity cues back into the global representations, effectively reversing the dilution process. Third, to overcome the absence of explicit entity-level supervision in traditional global contrastive objectives, we propose the fine-grained entity contrastive loss, a sample-level formulation that provides direct supervision for explicit entity alignment without requiring manual positive–negative pair annotation. Experiments on RSITMD, RSICD, and UCM-Captions datasets demonstrate that BEAN achieves leading mR scores of 50.74%, 34.67%, and 60.56%, respectively, consistently outperforming both PEFT-based methods and the fully fine-tuned CLIP baseline across all three benchmarks. The code is available at https://github.com/JasonGravity/BEAN**

***Index Terms*—Cross-modal alignment, image–text retrieval, parameter-efficient fine-tuning (PEFT), remote sensing (RS), vision-language models (VLMs).**

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I. Introduction

VISION-LANGUAGE models (VLMs) have revolutionized multimodal representation learning by demonstrating remarkable capabilities in bridging visual perception and linguistic semantics [1], [2]. Within the specialized domain of remote sensing (RS), this revolution has propelled cross-modal image–text retrieval into a highly impactful research frontier aimed at closing the semantic gap between complex overhead images and natural language descriptions [3]. Specifically, this field encompasses two bidirectional tasks: text-to-image (T2I) retrieval for locating target geographical scenes via textual descriptions, and image-to-text (I2T) retrieval for assigning precise linguistic summaries to Earth observation data. Consequently, robust performance across these retrieval tasks serves as the cornerstone for critical downstream applications such as interactive urban scene understanding [4], intelligent disaster response [5], and precision environmental monitoring [6].

To achieve this alignment, the architecture of image–text retrieval has evolved significantly. Early approaches rely on convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to project images and sentences into a shared latent space for coarse global alignment [7], [8]. Subsequently, attention mechanisms and Transformer architectures [9] enable fine-grained region–word alignment [10], [11], yet they often incur massive computational overhead. Recently, large-scale VLMs like contrastive language-image pretraining (CLIP) [1] have revolutionized the field by firmly establishing the pretraining and fine-tuning paradigm. By leveraging massive web-scale corpora, these foundational models construct a highly generalized open-vocabulary semantic space [12]. Within this paradigm, conventional adaptation to downstream tasks typically relies on full-parameter fine-tuning. However, because annotated RS datasets are significantly smaller than pretraining corpora, updating all parameters renders the network susceptible to severe overfitting and catastrophic forgetting, which degrades its inherent generalization [13]. To overcome these barriers, parameter-efficient fine-tuning (PEFT) has emerged as an advanced alternative. PEFT strategies freeze the majority of the pretrained backbone and introduce only a minuscule fraction of trainable parameters [14], [15]. This enables the model to efficiently absorb domain-specific RS knowledge while preserving pretrained representations and mitigating overfitting. Therefore, the synergy of frozen VLMs and PEFT

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modules has become the prevailing paradigm for cross-modal retrieval [16].

Despite the significant progress achieved by fine-tuning VLMs like CLIP, their underlying feature representation paradigm still suffers from fundamental limitations. Directly adapting these global-centric architectures to the highly complex RS domain exposes three critical structural and optimization bottlenecks.

1) *Semantic Ambiguity in Extracted Entity Representations:* Existing attempts to mitigate global dilution typically extract local features to represent visual and textual entities [3], [17]. However, these extracted representations are merely unguided feature clusters with highly ambiguous semantic References, inherently lacking explicit cross-modal grounding. Therefore, this semantic ambiguity severely hinders explicit entity alignment between modalities.
2) *Semantic Dilution of Fine-Grained Entities:* Existing dual-tower frameworks compress entire images and texts into singular global vectors (e.g., `[CLS]` and `[EOT]` tokens) [4], [6], optimizing solely for holistic similarity. However, highly homogeneous backgrounds dominate the representational space. Consequently, the semantic proportion of localized entities drops drastically, inevitably diluting critical fine-grained signals. This semantic imbalance renders the model insensitive to decisive local differences, triggering semantic ambiguity and severe retrieval degradation.
3) *Supervision Deficit and Annotation Bottleneck in Contrastive Learning:* Existing contrastive paradigms attempt to optimize cross-modal representations through holistic global objectives [4] and by explicitly constructing fine-grained positive/negative pairs [18]. However, global objectives fail to provide direct supervision for one-to-one entity correspondence, while traditional entity-level contrastive learning heavily relies on cumbersome annotation procedures. Hence, fine-grained optimization is trapped between ineffective indirect supervision and prohibitive annotation costs, severely bottlenecking accurate cross-modal alignment.

To address these challenges, we propose a bidirectional entity-level alignment network (BEAN), a PEFT-based dual-tower framework for fine-grained RS image–text retrieval. The contributions of this work are threefold, each directly targeting one of the identified limitations.

1) *Visual Entity Bank Constructor (VEBC) and Noun-to-Token Index Constructor (NTIC):* To resolve the severe semantic ambiguity and lack of explicit cross-modal grounding in existing entity representations, we introduce VEBC and NTIC as two complementary extraction modules. VEBC constructs a depth-stratified visual bank from entity patches, while NTIC utilizes precise noun-to-token mapping to extract the corresponding textual entity features, thereby generating semantically explicit representations across both modalities. By explicitly anchoring these dual-modal features, these modules transform unguided feature clusters into deterministically paired entities, successfully establishing the exact structural correspondence required for reliable fine-grained alignment.
2) *Cross-Modal Entity-Bidirectional Adapter (CEBA):* To counteract the severe semantic dilution of local entities during global pooling, we introduce the CEBA. Integrated via residual connections, CEBA executes a bidirectional feature routing that first contextualizes local entities with holistic scene information, and then reinjects these enriched representations back into the main sequence. By explicitly amplifying the semantic weight of entity tokens, this adapter successfully fortifies critical fine-grained cues against background dilution, while preserving the pretrained CLIP knowledge.
3) *Fine-Grained Entity Contrastive Loss:* To resolve the lack of fine-grained supervision and cumbersome annotation procedures, we propose the fine-grained entity contrastive loss. By leveraging the deterministically paired entities from VEBC and NTIC, this objective dynamically constructs positive and negative targets strictly within individual samples. Thus, it provides explicit entity-level supervision and shatters the manual annotation bottleneck to achieve accurate cross-modal fine-grained alignment.

The remainder of this article is organized as follows. Section II reviews related work. Section III details the methodology of our proposed framework. Section IV presents the experimental results. Finally, Section V concludes this article.

II. Related Work

In this section, related works in two main topics are introduced: PEFT in VLMs and cross-modal fine-grained alignment.

A. PEFT in VLMs

PEFT adapts large-scale VLMs to downstream tasks while circumventing the prohibitive computational costs of full-parameter updates. Existing PEFT methods fall into three primary categories.

1) Prompt-based tuning prepends learnable continuous vectors to steer the frozen backbone's representation space. CoOp [19] optimizes context vectors for the CLIP text encoder.
2) Adapter-based tuning inserts trainable modules into frozen Transformer layers. For instance, adapter [14] introduces a residual bottleneck structure (down-projection, nonlinearity, and up-projection), which AdaptFormer [20] extends to vision Transformers, and CLIP-adapter [16] applies to frozen CLIP encoders.
3) Reparameterization-based tuning constructs weight updates via low-rank decomposition matrices to eliminate inference latency, as pioneered by LoRA [15]. Notably, He et al. [21] demonstrate that these approaches share a unified mathematical framework, all interpretable as modifying hidden representations through residual vectors.

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In the RS cross-modal retrieval domain, these PEFT paradigms adapt VLMs to Earth observation imagery.

1) Prompt-based methods tailor the input space for domain-specific semantics. For instance, CUP [22] optimizes context- and uncertainty-aware prompts to bridge the modality gap, and SWPE [23] combines strong and weak prompts to capture fine-grained details and global semantics.
2) Adapter-based methods calibrate intermediate representations via trainable modules. Specifically, Yuan et al. [24] propose a parameter-efficient transfer learning framework with a multimodal RS adapter and a hybrid multimodal contrastive learning objective; CMER [25] reduces training memory consumption by introducing a side-branch focus-adapter; and HarMA [26] integrates a multimodal gated adapter to explicitly alleviate the excessive clustering of intramodal embeddings.
3) Reparameterization-based tuning leverages low-rank matrices to achieve efficient cross-modal alignment. For example, Yang et al. [27] propose PERT-RaMa to introduce Kronecker-product low-rank adaptations, effectively optimizing retrieval performance with minimal parameter updates.

Current RS-specific PEFT methods predominantly optimize the global feature stream, lacking explicit mechanisms to preserve local entity signals. Consequently, dominant background contexts easily dilute fine-grained entity cues, restricting precise cross-modal alignment. To address this, we establish explicit bidirectional interactions between the global stream and localized entity representations. Embedded directly within Transformer layers, our method highlights essential entity cues against background interference while keeping the pretrained backbone entirely frozen.

B. Cross-Modal Fine-Grained Alignment

Cross-modal fine-grained alignment aims to establish explicit correspondences between visual regions and textual entities. Early paradigms rely heavily on off-the-shelf object detectors. For instance, SCAN [10] computes fragment-level similarities via cross-attention, while SGRAF [28] relies on similarity graph reasoning and Oscar [29] leverages scene graphs and object tags. However, these methods suffer from heavy computational overhead and restricted bounding-box vocabularies. To bypass detection bottlenecks, subsequent approaches align image patches directly with textual tokens. ViLT [11] utilizes deep cross-modal transformers for dense interactions. Later, contrastive learning enabled fine-grained alignment without spatial annotations: FILIP [30] introduces tokenwise maximum similarity, while RegionCLIP [31] and GLIP [32] extend contrastive objectives to the region level. Most recently, models like LongCLIP [33], FineLIP [34], and FG-CLIP [35] enhance CLIP's fine-grained perception through extended text inputs, dynamic token aggregation, and large-scale fine-grained supervision, respectively.

In the RS cross-modal retrieval domain, fine-grained alignment has attracted substantial research effort. Recent RS image understanding studies have highlighted the importance of local spatial information through content-adaptive multiregion modeling and superpixel–pixel graph learning [36], [37]. Driven by the recognition that retrieval accuracy hinges not on holistic scene matching but on the precise correspondence between specific local entities and their textual descriptions, GaLR [17] constructs global–local feature representations and leverages graph reasoning to model region–word relationships, enabling alignment at multiple semantic granularities in aerial scenes. CPPMN [38] advances this by transforming implicit multimodal features into explicit relationship graphs, enabling dynamic alignment across semantic scales from global scenes to localized objects. FGVLA [18] further tightens local correspondences through spatial mask loss and fine-grained contrastive objectives, establishing explicit mappings between visual patches and textual nouns. Recently, Li et al. [39] push this line toward explicit entity-level alignment via a prompt-guided adapter mechanism that calibrates localized multimodal representations for fine-grained RS scene understanding.

Despite recent advances, fine-grained entity representations in existing frameworks continue to suffer from semantic ambiguity, lacking both explicit cross-modal correspondence and direct fine-grained loss supervision. To overcome these bottlenecks, we establish explicit cross-modal mappings of extracted entities with clear semantic relationships and introduce direct fine-grained supervision.

III. Methodology

Existing dual-tower models [1], [40] often overlook entity-level cues in fine-grained RS retrieval. To address this limitation, we propose the BEAN, a PEFT framework that freezes the pretrained backbones and updates only the adapter modules and projection layers. Specifically, Section III-A provides the overall architecture of the proposed BEAN. Next, Section III-B details the symmetric visual and textual entity construction, introducing both the VEBC and the NTIC. Following this, Section III-C presents the CEBA. Finally, Section III-D details the fine-grained entity contrastive loss.

A. Overall Architecture of BEAN

As illustrated in Fig. 1, BEAN takes an image–text pair (I, T) as input. During training, two preprocessing steps produce the entity-level priors. The spaCy parses K key nouns $N = \{n_1, \ldots, n_K\}$ from caption T, where K is sample-dependent and determined by the number of unique nouns in each caption. If the same noun is extracted multiple times from a caption, it is deduplicated and retained only once. The nouns then serve as open-vocabulary prompts to guide GroundingDINO [41] in cropping the corresponding localized patches $P = \{p_1, \ldots, p_K\}$ from image I along with their normalized bounding box coordinates $\mathbf{C} \in \mathbb{R}^{K\times 4}$. If multiple candidate patches are obtained for the same noun, only the highest confidence crop is retained. This constructs a one-to-one correspondence $(n_k \leftrightarrow p_k)$ at the data level. During inference, the image and text branches are processed independently. In the image branch, we employ the publicly available pretrained RAM++ model [42] to predict semantic tags from the input image I. spaCy then filters these tags to

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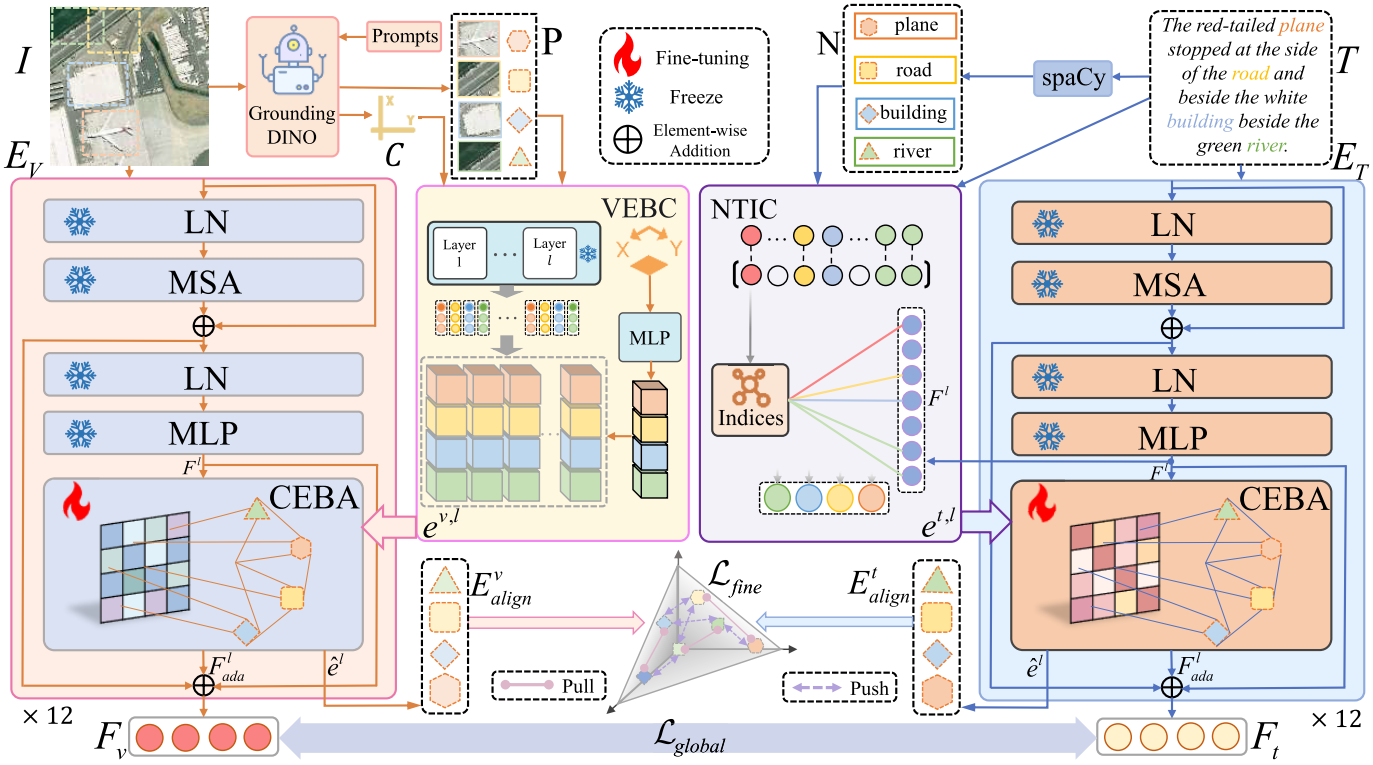


Fig. 1. Overview of the proposed BEAN framework. Here, VEBC, NTIC, and CEBA represent the VEBC, the NTIC, and the CEBA, respectively; MSA and LN denote multihead self-attention and layer normalization. Here, "Prompts" denote nouns extracted from the paired caption during training and nouns generated by RAM++ during inference.

retain noun-bearing categories, which are used as prompts for GroundingDINO to obtain the localized patches P and their coordinates $\mathbf{C}$. The text branch follows the same procedure as during training, where spaCy extracts entity nouns from each input text and NTIC identifies their corresponding token positions.

VEBC and NTIC then process these priors in parallel to produce layerwise entity representations. On the visual side, the VEBC takes the localized patches P and their spatial coordinates $\mathbf{C}$ as inputs to formulate the layerwise visual entity tokens for the lth layer

$$e^{v,l} = \text{VEBC}\,(P, C) \in \mathbb{R}^{K\times D} \tag{1}$$

where D denotes the hidden dimension.

On the textual side, the NTIC takes the full caption T, the extracted nouns N, and the intermediate hidden states F^l as inputs to generate the corresponding layerwise textual entity tokens

$$e^{t,l} = \text{NTIC}\left(T, N, F^l\right) \in \mathbb{R}^{K\times D} \tag{2}$$

where $F^l \in \mathbb{R}^{M\times D}$ (with M denoting the sequence length of the global tokens) is the global hidden states directly outputted by the MLP sublayer of the lth block of the frozen encoder.

To bridge the fine-grained entity tokens with the holistic global representations, CEBA is integrated via a residual connection after the MLP sublayer across all 12 Transformer layers in both E_V and E_T, serving as trainable parameters within BEAN. At each layer l, the CEBA block formulates the enhanced global hidden states and the updated entity representations as follows:

$$\left(F^l_{ada}, \hat{e}^l\right) = \text{CEBA}\left(F^l, e^l\right) \tag{3}$$

where $e^l \in \mathbb{R}^{K\times D}$ (with $e^l \in \{e^{v,l}, e^{t,l}\}$) represents the modality-specific entity tokens. The outputs consist of $F^l_{ada} \in \mathbb{R}^{M\times D}$, representing the enhanced global hidden states of the current layer, and $\hat{e}^l \in \mathbb{R}^{K\times D}$, which carries the enriched entity features.

After propagating through all 12 layers, the overall forward passes of the dual-tower encoders are formulated as

$$\begin{aligned}\left(F_v, E^v_{align}\right) &= E_V\left(I, \{e^{v,l}\}_{l=1}^{12}\right)\\ \left(F_t, E^t_{align}\right) &= E_T\left(T, \{e^{t,l}\}_{l=1}^{12}\right)\end{aligned} \tag{4}$$

where $F_v, F_t \in \mathbb{R}^D$ are the resulting global embeddings, and $E^v_{align}, E^t_{align} \in \mathbb{R}^{K\times D}$ are the final entity-level alignment sets in the twelfth Transformer block.

The BEAN is jointly optimized by a dual-level objective. The global embeddings (F_v, F_t) are supervised by a batch-level InfoNCE contrastive loss $\mathcal{L}_{global}$, while the entity alignment sets $(E^v_{align}, E^t_{align})$ are supervised by the fine-grained entity contrastive loss $\mathcal{L}_{fine}$. The total training objective is

$$\mathcal{L} = \mathcal{L}_{global} + \alpha \cdot \mathcal{L}_{fine} \tag{5}$$

where α is a scalar weight balancing the two supervision signals. A detailed sensitivity analysis of this hyperparameter is provided in Section IV-E.

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Algorithm 1 Training Procedure of BEAN

Input : Batched image-text pairs (I, T);
Localized visual patches P and spatial coordinates $\mathbf{C}$ parsed by GroundingDINO;
Nouns N extracted from captions T by spaCy.
Output : Optimal learnable parameters Θ_{train}.
Initialize: Freeze primary base weights of E_V, E_T
foreach *mini-batch* **do**
// **Step 1: Visual Entity Bank Construction**
Generate layer-wise visual entity tokens via VEBC:
$\{e^{v,l}\}_{l=1}^{12} \leftarrow \text{VEBC}(P, \mathbf{C})$;
// **Step 2: Dual-Tower Model Forward Pass**
Run image encoder to obtain global scene and local entity alignment:
$(F_v, E^v_{align}) \leftarrow E_V\left(I, \{e^{v,l}\}_{l=1}^{12}\right)$;
Run text encoder to obtain global scene and local entity alignment:
$(F_t, E^t_{align}) \leftarrow E_T\left(T, \{e^{t,l}\}_{l=1}^{12}\right)$;
// Note: Cross-modal Entity-Bidirectional Adapter (CEBA) is embedded within both E_V and E_T for layer-wise enhancement. Inside E_T, NTIC dynamically extracts features: $e^{t,l} \leftarrow \text{NTIC}(T, N, F^l)$
// **Step 3: Joint Dual-Level Optimization**
Compute global scene contrastive loss:
$\mathcal{L}_{global} \leftarrow \text{InfoNCE}(F_v, F_t, \tau)$;
Compute sample-level fine-grained entity contrastive loss:
$\mathcal{L}_{fine} \leftarrow \text{FineEntityLoss}(E^v_{align}, E^t_{align}, \tau)$;
Combine to compute total training objective:
$\mathcal{L} \leftarrow \mathcal{L}_{global} + \alpha \cdot \mathcal{L}_{fine}$;
Update parameters Θ_{train} via Backpropagation:
$\Theta_{\text{train}} \leftarrow \Theta_{\text{train}} - \eta \nabla_{\Theta_{\text{train}}} \mathcal{L}$;

The BEAN establishes a unified fine-grained entity alignment paradigm by extracting localized visual–textual entity features and bidirectionally fusing them with global representations, following the training procedure in Algorithm 1. Through the joint optimization of global and fine-grained entity-level losses, this dual-path design achieves comprehensive cross-modal semantic alignment at both scene and object levels.

B. Symmetric Visual and Textual Entity Construction

Existing approaches for local entity feature extraction typically suffer from ambiguous semantic representations. To address these limitations, we propose two modality-specific modules: the VEBC (VEBC: Fig. 2) and the NTIC (NTIC: Fig. 3). Specifically, VEBC constructs a visual entity bank by encoding and fusing preextracted patch features P with their projected spatial coordinate embeddings C, whereas the NTIC extracts a textual entity stream by slicing text encoder hidden states F^l guided by deterministic noun index mappings.

1) Visual Entity Bank Constructor: For visual entity streams, the VEBC module is designed to layerwise extract entity features from the frozen visual backbone and explicitly fuse them with absolute spatial coordinates. Specifically, VEBC takes a set of K preextracted entity image patches,

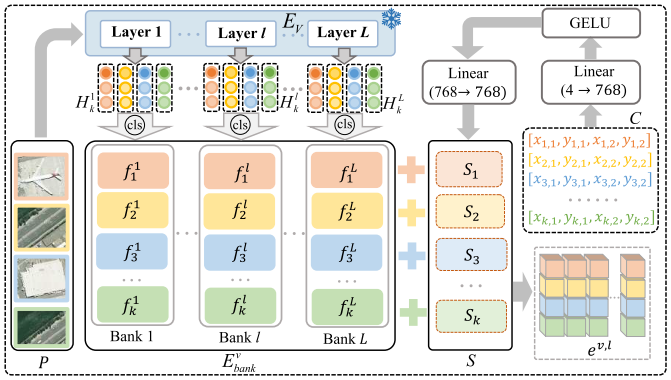


Fig. 2. Detailed illustration of the VEBC. Here, cls represents the [CLS] token extraction, and GELU serves as an activation function to introduce nonlinearity.

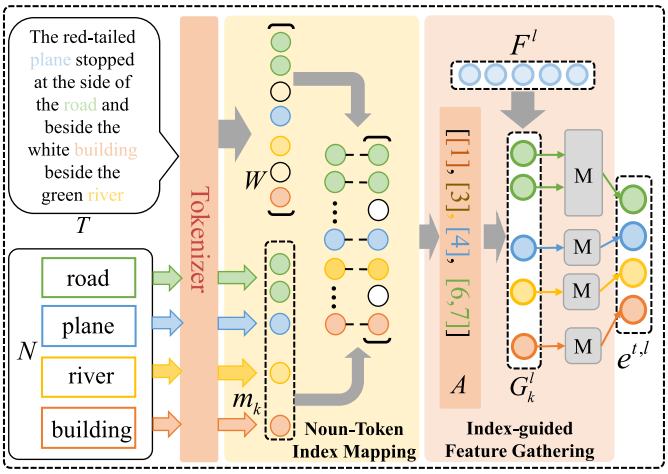


Fig. 3. Detailed illustration of the NTIC. Here, M represents the mean pooling.

$P = \{P_1, P_2, \ldots, P_K\}$, as one of its inputs. At each Transformer layer l ($l \in \{1, 2, \ldots, L\}$), the strictly frozen primary visual backbone E_V processes the K patches. Computed offline, the VEBC module functions as a dedicated feature extraction pathway that directly reuses E_V and bypasses the trainable CEBA modules to generate independent dense token sequences denoted as $\{\mathbf{H}^l_k\}_{k=1}^K$. Here, each sequence $\mathbf{H}^l_k \in \mathbb{R}^{N\times D}$ corresponds to the kth entity patch P_k, where N denotes the spatial sequence length (including the class token). To obtain highly condensed semantic representations, we extract the [CLS] token from each output sequence, yielding the decoupled visual feature vector $f^l_k = \mathbf{H}^l_k[0] \in \mathbb{R}^D$ for the kth entity at the lth layer. By collecting these tokens across all K entities at a specific layer l, we first construct a layer-specific entity bank, denoted as Bank l. We then stack these individual banks along the depth dimension to form a cross-layer visual semantic bank $E^v_{bank} \in \mathbb{R}^{L\times K\times D}$

$$E^v_{bank} = \{\text{Bank } l\}_{l=1}^L = \left\{\left\{f^l_k\right\}_{k=1}^K\right\}_{l=1}^L. \tag{6}$$

Although E^v_{bank} isolates the multilevel semantic identities of all entities, these features remain agnostic to their original physical locations within the global scene. To bridge the

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spatial gap, we introduce a continuous coordinate embedding mechanism. Let $C \in \mathbb{R}^{K\times 4}$ be the bounding box coordinates for all K entities. For the kth entity, its coordinate vector $C_k = [x_{k,1}, y_{k,1}, x_{k,2}, y_{k,2}] \in \mathbb{R}^4$ is projected into a spatial embedding $S_k \in \mathbb{R}^D$ via a two-layer nonlinear projection network

$$S_k = \mathbf{W}_{c2}\left(\mathrm{GELU}\left(\mathbf{W}_{c1}C_k + \mathbf{b}_{c1}\right)\right) + \mathbf{b}_{c2} \tag{7}$$

where $\mathbf{W}_{c1} \in \mathbb{R}^{D\times 4}$ and $\mathbf{W}_{c2} \in \mathbb{R}^{D\times D}$ are learnable projection weights, and $\mathbf{b}_{c1}$ and $\mathbf{b}_{c2}$ are the corresponding biases. This mapping constructs a layer-agnostic spatial embedding repository $S = \{S_1, \ldots, S_k, \ldots, S_K\} \in \mathbb{R}^{K\times D}$. Ultimately, the spatial embedding repository S is fused with the visual semantic bank E^v_{bank}. Because the physical location of an entity is invariant across network layers, the 2-D spatial matrix S is broadcast and added elementwise to the 3-D semantic bank E^v_{bank}. For any layer l, the spatially aware representation $e^{v,l}_k$ for the kth entity is given by

$$e^{v,l}_k = f^l_k + S_k. \tag{8}$$

This yields the final coordinate-injected visual entity tensor $e^{v,l} = \{e^{v,l}_1, e^{v,l}_2, \ldots, e^{v,l}_K\} \in \mathbb{R}^{K\times D}$ for the lth layer.

The VEBC systematically extracts and aggregates fine-grained visual features across all network layers. By independently encoding predefined image patches and injecting spatial coordinates, it constructs a multilevel entity bank. Accordingly, VEBC provides the spatially anchored entity stream required for the subsequent bidirectional integration within the CEBA module.

2) Noun-to-Token Index Constructor: For text entity streams, we construct the NTIC module, as illustrated in Fig. 3, to dynamically extract and aggregate localized semantic features from the intermediate hidden states of the text encoder E_T. Both the raw caption T and the preextracted core noun set $N = \{n_1, n_2, \ldots, n_K\}$ are concurrently fed into a standard subword tokenizer. Consequently, the caption T is converted into a global discrete token sequence W of length M, while each individual noun entity $n_k \in N$ is tokenized into a local subword sequence m_k. To localize these semantic concepts within the global sentence, NTIC executes a noun–token index mapping operation. By performing exact string-matching of the tokenized sequence m_k against the global token sequence W, we determine its coordinates and construct an index set $A_k \subset \{1, 2, \ldots, M\}$. Due to subword tokenization, a single noun is frequently partitioned into multiple consecutive tokens. The index set A_k accommodates a variable number of positional indices, recording the full sequence footprint of the kth entity within W. By executing this matching across all K nouns, we formulate the global noun–token index set $A = \{A_1, A_2, \ldots, A_K\}$, which serves as the structural guidance for feature extraction.

At the lth layer of the text encoder, guided by the global index set A, the corresponding hidden states are selectively extracted from the intermediate hidden states $F^l \in \mathbb{R}^{M\times D}$. Formally, for the kth entity, using its specific coordinate subset $A_k \in A$, the gathered raw textual feature matrix G^l_k is formulated as

$$G^l_k = F^l[A_k, :] \in \mathbb{R}^{|A_k|\times D} \tag{9}$$

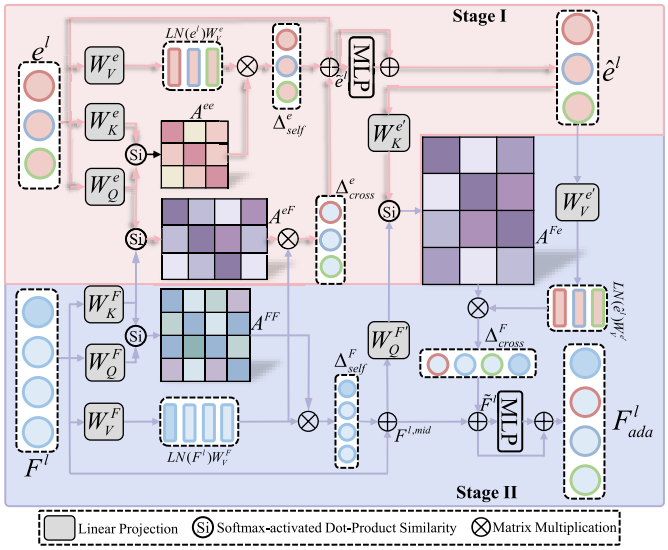


Fig. 4. Detailed architecture of the proposed CEBA block at layer l.

where $F^l[i, :] \in \mathbb{R}^D$ indicates the feature vector located at the ith position, and $|A_k|$ denotes the cardinality of the index set A_k. To condense these variable-length token features into a uniform representation, we apply a mean pooling operation across the discrete subword dimension. For the kth entity, the condensed textual semantic feature $e^{t,l}_k \in \mathbb{R}^D$ is computed as

$$e^{t,l}_k = \frac{1}{|A_k|}\sum_{j=1}^{|A_k|} G^l_k[j, :]. \tag{10}$$

Consequently, the NTIC module yields the final textual entity representation tensor $e^{t,l} = \{e^{t,l}_1, e^{t,l}_2, \ldots, e^{t,l}_K\} \in \mathbb{R}^{K\times D}$ for the lth layer. This output symmetrically mirrors the visual entity stream $e^{v,l}$, satisfying the structural prerequisites for the subsequent CEBA mechanism.

The NTIC establishes deterministic mappings between core nouns and their tokenized subwords to form a precise structural index. Guided by this index, it extracts and aggregates the corresponding hidden states from the pretrained text encoder to generate a structurally aligned textual entity stream required for subsequent bidirectional integration.

C. Cross-Modal Entity-Bidirectional Adapter

Standard dual-tower encoders inevitably dilute fine-grained entity signals when compressing all spatial tokens into a singular global embedding. To counteract this, we propose the CEBA, as illustrated in Fig. 4. By establishing a layer-wise bidirectional information flow, CEBA effectively prevents entity dilution through two stages. In Stage I, the isolated entity tokens proactively gather holistic scene contexts from the global stream. In Stage II, the global representations absorb these enhanced entity features to reinforce the fine-grained entity information within the global stream.

Stage I (Entity Enrichment): In this initial phase, the isolated entity representations e^l dynamically aggregate both their internal correlations and the holistic context from the global stream. To be specific, the entity stream first computes the

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intraentity attention map $A^{ee} \in \mathbb{R}^{K\times K}$ and its corresponding self-attention increment $\Delta^{e}_{\text{self}} \in \mathbb{R}^{K\times D}$ to model the geometric or semantic relationships among the K entities

$$A^{ee} = \text{Softmax}\left(\frac{\left(\text{LN}\left(e^{l}\right) W^{e}_{Q}\right)\left(\text{LN}\left(e^{l}\right) W^{e}_{K}\right)^{\top}}{\sqrt{d}}\right) \tag{11}$$

$$\Delta^{e}_{self} = A^{ee}\left(\text{LN}\left(e^{l}\right) W^{e}_{V}\right) \tag{12}$$

where LN(·) denotes the layer normalization operator, and Softmax(·) functions as the normalization operator to convert dot-product similarities into valid attention weights. Concurrently, to break the spatial isolation and selectively extract holistic semantic cues highly relevant to each specific entity from the global stream, the entity tokens act as queries to actively attend to the global states F^{l}. This process generates the cross-stream attention map $A^{eF} \in \mathbb{R}^{K\times M}$ and the context-aware cross-attention increment $\Delta^{e}_{\text{cross}} \in \mathbb{R}^{K\times D}$

$$A^{eF} = \text{Softmax}\left(\frac{\left(\text{LN}\left(e^{l}\right) W^{e}_{Q}\right)\left(\text{LN}\left(F^{l}\right) W^{F}_{K}\right)^{\top}}{\sqrt{d}}\right) \tag{13}$$

$$\Delta^{e}_{cross} = A^{eF}\left(\text{LN}\left(F^{l}\right) W^{F}_{V}\right) \tag{14}$$

where $W^{e}_{Q}, W^{e}_{K}, W^{e}_{V} \in \mathbb{R}^{D\times D}$ denote the learnable projection matrices for the entity stream, and $W^{F}_{K}, W^{F}_{V} \in \mathbb{R}^{D\times D}$ are the corresponding key and value projections from the global stream. The term $d = D/H$ represents the channel dimension of each attention head (with H being the number of heads), acting as a scaling factor to prevent excessively large attention logits and mitigate softmax saturation.

To integrate these structural and contextual features, we employ trainable gating scalars $\gamma^{e}_{self}, \gamma^{e}_{cross} \in \mathbb{R}$ to modulate the residual connections. The intermediate fused entity state $\tilde{e}^{l} \in \mathbb{R}^{K\times D}$ is obtained by

$$\tilde{e}^{l} = e^{l} + \gamma^{e}_{self}\Delta^{e}_{self} + \gamma^{e}_{cross}\Delta^{e}_{cross}. \tag{15}$$

Subsequently, to perform nonlinear semantic integration, $\tilde{e}^{l}$ is passed through a standard MLP. Modulated by a trainable gating parameter $\gamma^{e}_{\text{mlp}} \in \mathbb{R}$, this yields the enriched, context-aware entity representations $\hat{e}^{l} \in \mathbb{R}^{K\times D}$

$$\hat{e}^{l} = \tilde{e}^{l} + \gamma^{e}_{mlp}\text{MLP}\left(\text{LN}\left(\tilde{e}^{l}\right)\right). \tag{16}$$

This yields globally aware entity tokens $\hat{e}^{l}$ ready for the subsequent injection into the global stream.

Stage II (Global Injection): Once the entity representations are enriched, the global sequence initiates the second phase to absorb these enhanced local features. First, the global stream computes a self-attention map $A^{FF} \in \mathbb{R}^{M\times M}$ and its corresponding increment $\Delta^{F}_{\text{self}} \in \mathbb{R}^{M\times D}$ to capture macroscopic background dependencies

$$A^{FF} = \text{Softmax}\left(\frac{\left(\text{LN}\left(F^{l}\right) W^{F}_{Q}\right)\left(\text{LN}\left(F^{l}\right) W^{F}_{K}\right)^{\top}}{\sqrt{d}}\right) \tag{17}$$

$$\Delta^{F}_{self} = A^{FF}\left(\text{LN}\left(F^{l}\right) W^{F}_{V}\right) \tag{18}$$

where $W^{F}_{Q}, W^{F}_{K}, W^{F}_{V} \in \mathbb{R}^{D\times D}$ denote the query, key, and value projection matrices for the global stream, respectively. Next, this self-attention increment is integrated into the original global stream F^{l} via a gated residual connection controlled by a trainable scalar $\gamma^{F}_{self} \in \mathbb{R}$ to form the intermediate global state $F^{l,mid} \in \mathbb{R}^{M\times D}$

$$F^{l,mid} = F^{l} + \gamma^{F}_{self}\Delta^{F}_{self}. \tag{19}$$

Crucially, to explicitly inject the fine-grained entity features back into the global context, this intermediate state $F^{l,\text{mid}}$ acts as the query to extract targeted local semantics from the newly updated entity features $\hat{e}^{l}$. This process yields the reverse cross-attention map $A^{Fe} \in \mathbb{R}^{M\times K}$ and the cross-attention increment $\Delta^{F}_{cross} \in \mathbb{R}^{M\times D}$

$$A^{Fe} = \text{Softmax}\left(\frac{\left(\text{LN}\left(F^{l,mid}\right) W^{F\prime}_{Q}\right)\left(\text{LN}\left(\hat{e}^{l}\right) W^{e\prime}_{K}\right)^{\top}}{\sqrt{d}}\right) \tag{20}$$

$$\Delta^{F}_{cross} = A^{Fe}\left(\text{LN}\left(\hat{e}^{l}\right) W^{e\prime}_{V}\right) \tag{21}$$

where $W^{F\prime}_{Q}, W^{e\prime}_{K}, W^{e\prime}_{V} \in \mathbb{R}^{D\times D}$ denote the corresponding projection matrices for this reverse injection path. The cross-attention increment is then integrated to yield the combined global representation $\tilde{F}^{l} \in \mathbb{R}^{M\times D}$ via another trainable scalar $\gamma^{F}_{cross} \in \mathbb{R}$

$$\tilde{F}^{l} = F^{l,mid} + \gamma^{F}_{cross}\Delta^{F}_{cross}. \tag{22}$$

Ultimately, $\tilde{F}^{l}$ undergoes a nonlinear transformation via an MLP modulated by a trainable gating parameter $\gamma^{F}_{\text{mlp}} \in \mathbb{R}$ to consolidate the newly injected local semantics, yielding the adapted global representation $F^{l}_{\text{ada}} \in \mathbb{R}^{M\times D}$

$$F^{l}_{ada} = \tilde{F}^{l} + \gamma^{F}_{mlp}\text{MLP}\left(\text{LN}\left(\tilde{F}^{l}\right)\right). \tag{23}$$

The CEBA module serves as a bidirectional bridge that unifies macroscopic scene contexts with fine-grained local entity semantics. Through a two-stage mechanism, isolated entities first extract holistic feature information and subsequently project it inversely to explicitly enrich the global sequence. Ultimately, this architectural design effectively overcomes the local feature signal dilution bottleneck without compromising the pretrained foundation, yielding highly discriminative, entity-enhanced representations for the joint contrastive optimization task.

D. Fine-Grained Entity Contrastive Loss

Existing cross-modal contrastive objectives operate predominantly at the macro-level, encapsulating both the entire visual scene and its corresponding textual description into singular holistic representations. This paradigm inherently marginalizes local entity interactions. Contemporary local-alignment methods strive for fine-grained supervision but typically resort to either labor-intensive annotations of positive and negative pairs or computationally prohibitive runtime matching mechanisms. To overcome these bottlenecks, we propose the fine-grained entity contrastive loss ($\mathcal{L}_{fine}$).

The entity correspondence used in this loss is constructed during preprocessing. When the same noun occurs multiple times within a caption, only its first occurrence and the corresponding noun index are retained. If GroundingDINO produces multiple candidate boxes for the same noun, only the highest confidence valid crop is retained. Nouns without valid detections are discarded, and padded or invalid entity slots are excluded by the validity mask.

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By inheriting the deterministic one-to-one entity correspondence constructed in the preceding stages, the final visual and textual entity alignment sets ($E^{v}_{align}, E^{t}_{align} \in \mathbb{R}^{K\times D}$) are structurally aligned at the index level. For a given sample b, let $e^{v,b}_{k}, e^{t,b}_{k} \in \mathbb{R}^{D}$ be the kth individual feature vectors extracted from their respective alignment sets. Crucially, because each feature vector possesses an explicit semantic reference governed by the shared index k, cross-modal correspondence is inherently established. Consequently, identical indices $(e^{v,b}_{k}, e^{t,b}_{k})$ natively form ground-truth positive pairs, while distinct indices naturally construct negative pairs. This mechanism automatically generates fine-grained training samples, bypassing the need for labor-intensive annotations of positive and negative pairs.

Formally, for a sample b containing K valid entities, the intrasample fine-grained alignment loss $\mathcal{L}^{b}_{fine}$ adaptively transitions between a symmetric InfoNCE formulation and a cosine-distance fallback depending on the entity count K

$$\mathcal{L}^{b}_{fine} = \begin{cases} \frac{1}{2}\left(\mathcal{L}^{b}_{\text{v2t}} + \mathcal{L}^{b}_{\text{t2v}}\right), & \text{if } K \geq 2 \\ w \cdot \left(1 - \hat{\mathbf{e}}^{v,b}_{1} \cdot \hat{\mathbf{e}}^{t,b}_{1}\right), & \text{if } K = 1 \end{cases} \tag{24}$$

where $w = 0.5$ represents the scaling weight which compensates for the absence of contrastive variance.

For dense entity scenarios ($K \geq 2$), the visual-to-textual alignment loss $\mathcal{L}^{b}_{\text{v2t}}$ and the textual-to-visual alignment loss $\mathcal{L}^{b}_{\text{t2v}}$ are explicitly defined as

$$\mathcal{L}^{b}_{\text{v2t}} = -\frac{1}{K}\sum_{k=1}^{K} \log \frac{\exp\left(\hat{\mathbf{e}}^{v,b}_{k} \cdot \hat{\mathbf{e}}^{t,b}_{k} / \tau\right)}{\sum_{j=1}^{K} \exp\left(\hat{\mathbf{e}}^{v,b}_{k} \cdot \hat{\mathbf{e}}^{t,b}_{j} / \tau\right)} \tag{25}$$

$$\mathcal{L}^{b}_{\text{t2v}} = -\frac{1}{K}\sum_{k=1}^{K} \log \frac{\exp\left(\hat{\mathbf{e}}^{t,b}_{k} \cdot \hat{\mathbf{e}}^{v,b}_{k} / \tau\right)}{\sum_{j=1}^{K} \exp\left(\hat{\mathbf{e}}^{t,b}_{k} \cdot \hat{\mathbf{e}}^{v,b}_{j} / \tau\right)}. \tag{26}$$

Here, $\hat{\mathbf{e}}^{v,b}_{k} = \mathbf{e}^{v,b}_{k}/\|\mathbf{e}^{v,b}_{k}\|_2$ and $\hat{\mathbf{e}}^{t,b}_{k} = \mathbf{e}^{t,b}_{k}/\|\mathbf{e}^{t,b}_{k}\|_2$ denote the ℓ_2-normalized entity representations, while τ serves as a shared learnable temperature parameter.

To explicitly harness this structural alignment without introducing semantic collision, our InfoNCE formulation ($K \geq 2$) deliberately abandons the conventional batch-level contrastive paradigm. Flattening entities across an entire batch introduces severe false-negative gradient noise. For instance, identical concepts (e.g., a "plane") appearing in different images would be erroneously forced to repel one another. Accordingly, we strictly confine the contrastive negative set within each individual image–text sample (i.e., intrasample contrastive learning). For a given sample b, the target visual entity $\hat{\mathbf{e}}^{v,b}_{k}$ is exclusively compared against the intrascene textual entities $\{\hat{\mathbf{e}}^{t,b}_{j}\}^{K}_{j=1}$. Consequently, $(\hat{\mathbf{e}}^{v,b}_{k}, \hat{\mathbf{e}}^{t,b}_{k})$ acts as the deterministic positive pair, while all co-occurring off-diagonal entities ($j \neq k$) provide reliable, hard negative supervision.

Conversely, in sparse entity scenarios ($K = 1$), the intrasample negative set is inherently empty, rendering the InfoNCE formulation degenerate. To retain full supervision, the loss defaults to the cosine-distance objective.

TABLE I
BENCHMARK DATASETS

Attribute	RSITMD [3]	RSICD [43]	UCM-Cap [44]
Images	4,743	10,921	2,100
Image Size	256×256	224×224	256×256
Spatial Resolution	Variable	Variable	~0.3m
Categories	32	30	21
Captions	23,715	54,605	10,500

Finally, the total fine-grained entity contrastive loss is computed by averaging across all B samples within the current batch

$$\mathcal{L}_{fine} = \frac{1}{B}\sum_{b=1}^{B} \mathcal{L}^{b}_{fine} \tag{27}$$

where B denotes the total batch size.

Fine-grained entity contrastive loss ($\mathcal{L}_{fine}$) leverages explicit index correspondence to natively formulate contrastive pairs. This design not only precisely anchors exact cross-modal positives but also naturally transforms other intrascene entities into high-quality hard negatives. Concurrently, by strictly confining the contrastive scope to individual samples, this objective fundamentally eliminates cross-batch false negatives, providing the robust fine-grained supervision required for joint framework optimization.

IV. EXPERIMENTAL STUDY

A. Datasets and Metrics

We evaluate BEAN on three standard RSITR datasets, including RSITMD [3], RSICD [43], and UCM-Captions [44], as summarized in Table I. Following established protocols [3], [43], we report Recall@K (R@K, $K \in \{1, 5, 10\}$) for both I2T and T2I retrieval. We further use mean recall (mR) to measure overall retrieval performance, defined as the average of the six recall scores

$$\text{mR} = \frac{1}{6}\sum_{K\in\{1,5,10\}} \left(\text{R@K}_{\text{image}} + \text{R@K}_{\text{text}}\right). \tag{28}$$

B. Experimental Setting

All experiments are conducted on two NVIDIA V100 GPUs (32 GB each) using distributed data parallel (DDP) training. Each dataset is split into training, validation, and test subsets following the 80%/10%/10% protocol, and all input images are resized to 224×224. BEAN is built upon a frozen OpenCLIP ViT-B/32 backbone, and all backbone parameters are held fixed throughout training. The model is optimized for 15 epochs using AdamW with a batch size of 64, an initial learning rate of 2×10^{-4}, and a weight decay of 0.06. The learning rate follows a linear decay schedule after a warm-up phase covering the first 10% of training steps, and gradient norms are clipped to 1.0. The joint embedding dimension is $D = 512$. GroundingDINO is configured with a box threshold of 0.35 and a text threshold of 0.25.

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TABLE II

PERFORMANCE COMPARISON ON RSITMD DATASET. , , AND DENOTE THE FIRST, SECOND, AND THIRD BEST RESULTS

Methods	Publication	Backbone	Image Retrieval (%)			Text Retrieval (%)			mR (%)
			R@1	R@5	R@10	R@1	R@5	R@10	
Traditional-Based Full Training Methods									
VSE++ [8]	BMVC'18	ResNet18, GRU	7.79	24.87	38.67	10.38	27.65	39.60	24.83
SCAN t-i [10]	ECCV'18	Faster R-CNN, BiGRU	10.10	28.98	43.53	10.18	28.53	38.49	26.64
SCAN i-t [10]	ECCV'18	Faster R-CNN, BiGRU	9.82	29.38	42.12	11.06	25.88	39.38	26.28
MTFN [45]	ACM MM'19	Faster R-CNN, BiGRU	9.96	31.37	45.84	10.40	27.65	36.28	26.92
LW-MCR-u [46]	TGRS'22	SqueezeNet, CNN	9.25	34.07	54.03	9.73	26.77	37.61	28.58
AMFMN-sim [3]	TGRS'22	ResNet18, BiGRU	11.51	34.69	54.87	10.63	24.78	41.81	29.72
GaLR MR [17]	TGRS'22	ResNet18, BiGRU	11.15	36.68	51.68	14.82	31.64	42.48	31.41
PIR [47]	ACM MM'23	Swin-T+ResNet-50, BERT	12.17	41.68	63.41	18.14	41.15	52.88	38.24
Full fine-tuned CLIP [24]	TGRS'23	CLIP-ViT(ViT-B/32)	20.40	50.53	68.54	24.16	47.12	61.28	45.33
MSA [48]	TGRS'24	ResNet18, BERT	13.63	40.80	57.92	19.25	37.61	50.44	36.61
SMLGN [49]	TGRS'24	PVT, BERT	13.19	43.94	60.40	17.26	39.38	51.55	37.62
CMPAGL [50]	TGRS'24	Transformer, BERT	15.67	44.32	60.51	19.13	42.36	54.74	39.46
MSITA [51]	TGRS'24	Transformer, BiGRU	12.15	39.92	57.72	15.22	34.20	47.65	34.48
PGRN [52]	JSTAR'25	ResNet101, BERT	15.09	45.97	64.29	18.14	37.17	51.33	38.67
Wang et al. [53]	JSTAR'25	Swin Transformer, BERT	13.62	43.85	66.19	18.14	38.71	53.54	39.01
ESRN [54]	ICMR'25	Swin Transformer, Bi-GRU	16.37	48.14	66.19	18.58	42.70	57.30	41.55
Parameter-Efficient Fine-Tuning Methods									
Linear Probe [1]	ICML'21	CLIP-ViT(ViT-B/32)	13.33	41.80	63.89	17.02	33.12	48.35	36.25
CoOp [19]	IJCV'22	CLIP-ViT(ViT-B/32)	9.16	33.85	54.35	12.19	30.69	42.82	30.51
CoCoOp [55]	CVPR'22	CLIP-ViT(ViT-B/32)	10.21	34.20	53.36	14.53	32.89	45.94	31.85
AdaptFormer [20]	NeurIPS'22	CLIP-ViT(ViT-B/32)	14.27	41.53	61.46	16.71	30.16	42.91	34.50
BitFit [56]	ACL'22	CLIP-ViT(ViT-B/32)	14.93	42.37	62.05	15.66	35.13	48.41	36.43
MaPLE [57]	CVPR'23	CLIP-ViT(ViT-B/32)	13.59	43.25	61.92	16.96	39.97	53.02	38.12
PE-RSITR [24]	TGRS'23	CLIP-ViT(ViT-B/32)	20.10	50.63	67.97	23.67	44.07	60.36	44.47
CLIP-Adapter [16]	IJCV'24	CLIP-ViT(ViT-B/32)	13.30	40.20	60.06	12.83	28.84	39.05	32.38
AdaKron [58]	ACL'24	CLIP-ViT(ViT-B/32)	13.80	43.38	63.97	15.71	34.87	47.70	36.57
HarMA [26]	ICLRw'24	CLIP-ViT(ViT-B/32)	19.92	53.27	71.21	25.81	48.37	60.61	46.53
Uniadapter [59]	ICLR'24	CLIP-ViT(ViT-B/32)	17.54	44.89	65.46	19.86	36.32	51.28	39.23
LoKr [60]	ICLR'24	CLIP-ViT(ViT-B/32)	16.88	47.38	67.22	17.65	40.31	53.76	40.53
CUP [22]	TNN'24	CLIP-ViT(ViT-B/32)	17.70	48.40	64.82	21.57	43.70	56.75	42.16
Cross-Modal Adapter [61]	PR'25	CLIP-ViT(ViT-B/32)	16.31	44.33	64.75	18.16	36.08	48.72	38.06
PERT [27]	TGRS'25	CLIP-ViT(ViT-B/32)	17.95	50.81	69.08	20.66	44.16	58.94	43.60
FRS-Adapter [62]	JSTAR'25	CLIP-ViT(ViT-B/32)	20.44	50.70	68.25	23.81	44.53	60.47	44.70
FGVLA [18]	TGRS'25	CLIP-ViT(ViT-B/32)	21.77	54.12	72.92	26.33	51.99	64.38	48.58
CMER [25]	JAG'26	CLIP-ViT(ViT-B/32)	16.38	53.49	79.70	21.82	47.64	62.24	46.88
RRSITR [63]	arXiv'26	CLIP-ViT(ViT-B/32)	20.19	53.04	70.12	24.60	46.68	58.85	45.58
Ours									
BEAN	-	**CLIP-ViT(ViT-B/32)**	**24.20**	**57.39**	**74.82**	**28.76**	**53.54**	**65.71**	**50.74**

C. Comparison Studies

The quantitative results of BEAN and the competing methods on RSITMD, RSICD, and UCM-Captions are presented in Tables II–IV, respectively. The competing methods are organized into two groups: methods using full training or full fine-tuning, and PEFT methods based on CLIP ViT-B/32. Image retrieval refers to T2I retrieval, whereas text retrieval refers to I2T retrieval. All performance differences reported below are expressed in percentage points.

1) Results on RSITMD: As shown in Table II, BEAN achieves an mR of 50.74%, the highest among all compared methods. Among PEFT methods, BEAN surpasses FGVLA [18] (48.58%), CMER [25] (46.88%), HarMA [26] (46.53%), RRSITR [63] (45.58%), and FRS-Adapter [62] (44.70%) by 2.16, 3.86, 4.21, 5.16, and 6.04 percentage points, respectively. BEAN achieves the highest recall on five of the six individual retrieval metrics. In particular, its I2T R@1 reaches 28.76%, exceeding FGVLA (26.33%) and PE-RSITR [24] (23.67%) by 2.43 and 5.09 percentage points, respectively. For T2I retrieval, BEAN achieves R@1 and R@5 scores of 24.20% and 57.39%, improving upon FGVLA by 2.43 and 3.27 percentage points. Its T2I R@10 of 74.82% ranks second to CMER (79.70%). Compared with the fully fine-tuned CLIP baseline [24] (45.33% mR), BEAN improves mR by 5.41 percentage points. These results support the effectiveness of entity-aware adaptation for improving retrieval performance.

2) Results on RSICD: Table III shows that BEAN achieves an mR of 34.67%, the highest among the methods listed in the table. BEAN outperforms FGVLA [18] (34.12%), HarMA [26] (33.62%), RRSITR [63] (32.70%), and PE-RSITR [24] (31.12%) by 0.55, 1.05, 1.97, and 3.55 percentage points, respectively. For T2I retrieval, BEAN obtains the highest scores at all three recall levels, reaching 13.76%, 38.30%, and 53.92% for R@1, R@5, and R@10, respectively. For I2T retrieval, BEAN matches FGVLA at the highest R@1 of 16.93% and achieves the highest R@10 of 49.22%. Its I2T R@5 of 35.86% is slightly below FGVLA's 36.23%, indicating that the gains are not uniform across all metrics. BEAN also exceeds the fully fine-tuned CLIP baseline [24] (30.39% mR) by 4.28 percentage points, demonstrating the effectiveness of the proposed adaptation framework on RSICD.

3) Results on UCM-Captions: As shown in Table IV, BEAN achieves an mR of 60.56%, the highest among all compared methods. It surpasses FGVLA [18] (59.35%), PERT [27] (56.83%), and HarMA [26] (56.25%) by 1.21, 3.73, and 4.31 percentage points, respectively. The gains are particularly evident in T2I retrieval, where BEAN achieves the highest R@1, R@5, and R@10 scores of 24.19%, 70.76%, and 97.43%, respectively. Compared with FGVLA, these represent improvements of 4.48, 6.09, and 1.43 percentage points. For I2T retrieval, BEAN matches FGVLA at the highest R@1 of 24.29%, while its R@5 and R@10 scores of 63.81%

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TABLE III

PERFORMANCE COMPARISON ON RSICD DATASET. , , AND DENOTE THE FIRST, SECOND, AND THIRD BEST RESULTS

Methods	Publication	Backbone	Image Retrieval (%)			Text Retrieval (%)			mR (%)
			R@1	R@5	R@10	R@1	R@5	R@10	
Traditional-Based Full Training Methods									
VSE++ [8]	BMVC'18	ResNet18, GRU	2.82	11.32	18.10	3.38	9.51	17.46	10.43
SCAN t-i [10]	ECCV'18	Faster R-CNN, BiGRU	3.91	16.20	26.49	4.39	10.90	17.64	13.25
SCAN i-t [10]	ECCV'18	Faster R-CNN, BiGRU	3.71	16.40	26.73	5.85	12.89	19.84	14.23
MTFN [45]	ACM MM'19	Faster R-CNN, BiGRU	4.90	17.17	29.49	5.02	12.52	19.74	14.81
LW-MCR-u [46]	TGRS'22	SqueezeNet, CNN	4.30	18.85	32.34	4.39	13.35	20.29	15.59
AMFMN-sim [3]	TGRS'22	ResNet18, BiGRU	4.90	18.28	31.44	5.39	15.08	23.40	16.42
GaLR MR [17]	TGRS'22	ResNet18, BiGRU	4.69	19.48	32.13	6.59	19.85	31.04	18.96
PIR [47]	ACM MM'23	Swin-T+ResNet-50, BERT	6.97	24.56	38.92	9.88	27.26	39.16	24.46
Full fine-tuned CLIP [24]	TGRS'23	CLIP-ViT(ViT-B/32)	11.55	33.14	49.83	13.54	30.83	43.46	30.39
MSA [48]	TGRS'24	ResNet18, BERT	5.97	22.93	39.58	7.96	23.33	35.77	22.59
SMLGN [49]	TGRS'24	PVT, BERT	7.85	27.14	42.58	8.87	25.53	37.24	24.87
CMPAGL [50]	TGRS'24	Transformer, BERT	9.08	31.11	46.81	12.71	29.55	42.54	28.63
MSITA [51]	TGRS'24	Transformer, BiGRU	6.13	21.98	35.39	8.67	22.71	33.91	21.47
PGRN [52]	JSTAR'25	ResNet101, BERT	8.69	29.62	45.31	12.53	30.01	41.08	27.87
Wang et al. [53]	JSTAR'25	Swin Transformer, BERT	6.41	24.77	42.23	10.81	24.54	37.58	24.39
ESRN [54]	ICMR'25	Swin Transformer, Bi-GRU	9.80	30.71	46.31	12.76	30.74	43.78	29.01
Parameter-Efficient Fine-Tuning Methods									
Linear Probe [1]	ICML'21	CLIP-ViT(ViT-B/32)	7.81	25.89	42.47	8.46	24.41	37.72	24.46
CoOp [19]	IJCV'22	CLIP-ViT(ViT-B/32)	4.31	17.89	31.73	6.32	15.89	27.60	17.29
CoCoOp [55]	CVPR'22	CLIP-ViT(ViT-B/32)	6.15	21.68	34.29	7.65	24.25	36.08	21.68
AdaptFormer [20]	NeurIPS'22	CLIP-ViT(ViT-B/32)	9.09	29.89	46.81	12.46	28.49	41.86	28.10
BitFit [56]	ACL'22	CLIP-ViT(ViT-B/32)	8.28	28.11	43.97	10.32	24.74	37.13	25.42
MaPLE [57]	CVPR'23	CLIP-ViT(ViT-B/32)	6.88	24.47	40.12	7.93	26.78	42.42	24.77
PE-RSITR [24]	TGRS'23	CLIP-ViT(ViT-B/32)	11.63	33.92	50.73	14.13	31.51	44.78	31.12
CLIP-Adapter [16]	IJCV'24	CLIP-ViT(ViT-B/32)	7.67	24.87	39.73	7.11	19.48	31.01	21.65
AdaKron [58]	ACL'24	CLIP-ViT(ViT-B/32)	9.02	30.12	46.45	10.12	26.99	39.78	27.08
HarMA [26]	ICLRw'24	CLIP-ViT(ViT-B/32)	12.92	37.17	53.07	16.36	34.48	47.74	33.62
Uniadapter [59]	ICLR'24	CLIP-ViT(ViT-B/32)	9.61	30.06	47.16	12.65	30.81	42.74	28.84
LoKr [60]	ICLR'24	CLIP-ViT(ViT-B/32)	10.48	32.96	49.93	11.86	29.33	43.73	29.72
CUP [22]	TNN'24	CLIP-ViT(ViT-B/32)	9.86	31.60	47.15	12.05	32.66	48.70	30.34
Cross-Modal Adapter [61]	PR'25	CLIP-ViT(ViT-B/32)	9.57	30.74	48.36	11.18	27.31	40.62	27.96
PERT [27]	TGRS'25	CLIP-ViT(ViT-B/32)	11.47	34.04	51.46	12.97	31.53	45.20	31.11
FRS-Adapter [62]	JSTAR'25	CLIP-ViT(ViT-B/32)	11.72	34.11	50.96	14.22	31.57	44.93	31.25
FGVLA [18]	TGRS'25	CLIP-ViT(ViT-B/32)	12.72	36.61	53.76	16.93	36.23	48.49	34.12
CMER [25]	JAG'26	CLIP-ViT(ViT-B/32)	9.76	32.58	50.81	13.29	31.01	43.15	30.10
RRSITR [63]	arXiv'26	CLIP-ViT(ViT-B/32)	12.50	34.22	50.12	16.56	34.80	48.01	32.70
Ours									
BEAN	-	**CLIP-ViT(ViT-B/32)**	**13.76**	**38.30**	**53.92**	**16.93**	**35.86**	**49.22**	**34.67**

and 82.86% remain below FGVLA's 64.76% and 86.67%. Compared with PGRN [52] (52.72% mR) and the fully fine-tuned CLIP baseline [24] (52.43% mR), BEAN improves mR by 7.84 and 8.13 percentage points, respectively.

4) Overall Analysis: BEAN achieves the highest mR among the compared PEFT methods on all three benchmarks, with improvements of 2.16, 0.55, and 1.21 percentage points over FGVLA on RSITMD, RSICD, and UCM-Captions, respectively. These results support the effectiveness of combining entity feature extraction through VEBC and NTIC, bidirectional feature enhancement through CEBA, and explicit entity-level supervision through $\mathcal{L}_{\text{fine}}$. Although BEAN does not lead on every individual recall metric, it consistently improves overall retrieval performance across the three datasets. The componentwise contributions are further examined in the ablation study.

5) Qualitative Analysis: Fig. 5 presents top-5 retrieval examples on the RSITMD dataset. For image retrieval, the query describes a "small wild road" within "green farmland." The global-adaptation baselines, including CLIP and HarMA, tend to retrieve images that match the general farmland appearance but do not consistently preserve the road structure. In contrast, BEAN gives greater attention to the joint spatial and semantic configuration and retrieves the ground-truth image at Rank 1.

For text retrieval, the query image contains a tennis court together with surrounding plants and houses. Several baseline methods return captions that match only partial scene elements, such as trees or houses, while overlooking the tennis court. BEAN retrieves a caption that jointly describes the main entities and their surrounding context at Rank 1. These examples illustrate that VEBC, NTIC, and CEBA improve the preservation and integration of fine-grained entity information, leading to more precise cross-modal retrieval results.

D. Ablation Study and Evaluation

To evaluate the contribution of each proposed component, we perform an ablation study on all three benchmarks. The results are reported in Table V. We begin with a frozen CLIP ViT-B/32 baseline without adaptation modules and progressively add: 1) the visual entity path, which consists of CEBA(V) and VEBC (+C(V)+V); 2) the textual entity path, which consists of CEBA(T) and NTIC (+C(T)+N); and 3) the fine-grained entity contrastive loss (+$\mathcal{L}_{\text{fine}}$), resulting in the complete BEAN model.

1) Effect of CEBA(V)+VEBC: Adding the visual entity path improves mR by 0.93, 0.87, and 3.21 percentage points on RSITMD, RSICD, and UCM-Captions, respectively. These improvements indicate that incorporating entity-level visual information enhances the representation of fine-grained scene

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TABLE IV

PERFORMANCE COMPARISON ON UCM-CAPTIONS DATASET. , , AND DENOTE THE FIRST, SECOND, AND THIRD BEST RESULTS

Methods	Publication	Backbone	Image Retrieval (%)			Text Retrieval (%)			mR (%)
			R@1	R@5	R@10	R@1	R@5	R@10	
Traditional-Based Full Training Methods									
VSE++ [8]	BMVC'18	ResNet18, GRU	10.10	31.80	56.85	12.38	44.76	65.71	36.93
SCAN t-i [10]	ECCV'18	Faster R-CNN, BiGRU	12.76	50.38	77.24	14.29	45.71	67.62	44.67
SCAN i-t [10]	ECCV'18	Faster R-CNN, BiGRU	12.48	46.86	71.71	12.85	47.14	69.52	43.43
MTFN [45]	ACM MM'19	Faster R-CNN, BiGRU	14.19	52.38	78.95	10.47	47.62	64.29	44.65
LW-MCR-u [46]	TGRS'22	SqueezeNet, CNN	13.14	50.38	79.52	18.10	47.14	63.81	45.35
AMFMN-sim [3]	TGRS'22	ResNet18, BiGRU	12.86	53.24	79.43	16.67	45.71	68.57	46.08
Full fine-tuned CLIP [24]	TGRS'23	CLIP-ViT(ViT-B/32)	13.90	56.95	91.81	17.14	55.24	79.52	52.43
MSA [48]	TGRS'24	ResNet18, BERT	11.52	53.24	89.62	15.24	48.57	70.48	48.11
SMLGN [49]	TGRS'24	PVT, BERT	14.29	52.76	84.67	12.86	49.52	75.71	48.30
CMPAGL [50]	TGRS'24	Transformer, BERT	13.52	58.95	93.05	12.86	47.62	72.86	49.81
MSITA [51]	TGRS'24	Transformer, BiGRU	14.29	57.16	91.58	16.86	49.33	73.33	50.43
PGRN [52]	JSTAR'25	ResNet101, BERT	13.24	61.24	94.67	18.10	55.71	73.33	52.72
Parameter-Efficient Fine-Tuning Methods									
Linear Probe [1]	ICML'21	CLIP-ViT(ViT-B/32)	14.43	59.42	90.28	13.33	52.71	77.62	51.30
CoOp [19]	IJCV'22	CLIP-ViT(ViT-B/32)	9.78	48.34	87.41	7.19	42.76	74.19	44.95
CoCoOp [55]	CVPR'22	CLIP-ViT(ViT-B/32)	11.84	51.21	84.81	14.76	47.94	77.94	48.08
AdaptFormer [20]	NeurIPS'22	CLIP-ViT(ViT-B/32)	18.74	63.49	91.16	16.92	54.39	77.46	53.69
BitFit [56]	ACL'22	CLIP-ViT(ViT-B/32)	15.62	60.88	95.83	15.62	52.57	79.43	53.32
MaPLE [57]	CVPR'23	CLIP-ViT(ViT-B/32)	12.21	52.80	84.81	13.97	49.68	75.24	48.12
PE-RSITR [24]	TGRS'23	CLIP-ViT(ViT-B/32)	18.82	62.84	93.72	22.71	55.81	80.33	55.71
CLIP-Adapter [16]	IJCV'24	CLIP-ViT(ViT-B/32)	14.61	51.03	84.14	9.83	42.52	66.79	44.82
AdaKron [58]	ACL'24	CLIP-ViT(ViT-B/32)	13.39	55.73	89.52	14.57	49.81	73.90	49.49
HarMA [26]	ICLRw'24	CLIP-ViT(ViT-B/32)	19.81	65.33	95.71	22.38	57.14	77.14	56.25
Uniadapter [59]	ICLR'24	CLIP-ViT(ViT-B/32)	16.74	61.43	95.76	14.46	55.19	83.95	54.59
LoKr [60]	ICLR'24	CLIP-ViT(ViT-B/32)	17.12	64.21	95.83	19.33	58.10	82.29	56.15
CUP [22]	TNN'24	CLIP-ViT(ViT-B/32)	15.06	56.98	91.14	15.24	54.76	79.37	52.09
Cross-Modal Adapter [61]	PR'25	CLIP-ViT(ViT-B/32)	17.43	61.45	95.48	13.77	50.57	81.41	53.35
PERT [27]	TGRS'25	CLIP-ViT(ViT-B/32)	18.82	64.90	94.82	19.05	59.05	84.38	56.83
FRS-Adapter [62]	JSTAR'25	CLIP-ViT(ViT-B/32)	19.03	63.07	93.84	22.83	58.90	80.72	56.89
FGVLA [18]	TGRS'25	CLIP-ViT(ViT-B/32)	19.71	64.67	96.00	24.29	64.76	86.67	59.35
Ours									
BEAN	-	**CLIP-ViT(ViT-B/32)**	**24.19**	**70.76**	**97.43**	**24.29**	**63.81**	**82.86**	**60.56**

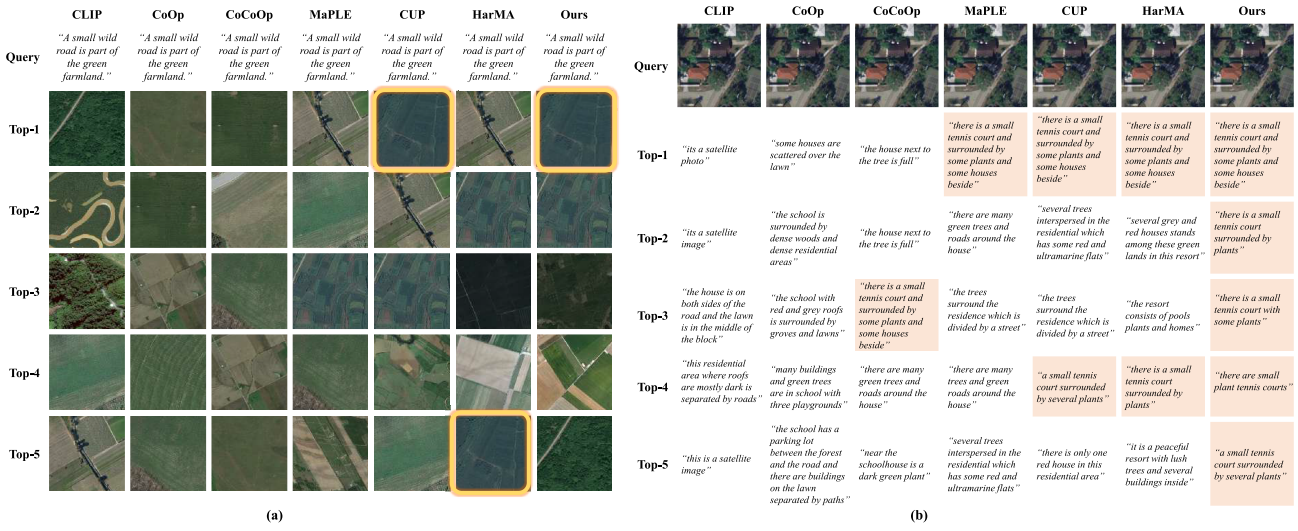


Fig. 5. Qualitative comparison of (a) image retrieval and (b) text retrieval results on the RSITMD dataset.

content. In particular, VEBC provides layerwise visual entity features together with spatial information, enabling the model to preserve local details that may be weakened in the global representation.

2) Effect of CEBA(T)+NTIC: Adding the textual entity path on top of the visual entity path further improves mR by 3.43, 3.01, and 3.32 percentage points on RSITMD, RSICD, and UCM-Captions, respectively. The larger gains suggest that explicit noun-level feature extraction is beneficial for these datasets. NTIC identifies the token positions associated with each noun and aggregates the corresponding hidden states into entity-specific representations. CEBA(T) then integrates these representations with the global textual stream, reducing the ambiguity caused by mixing entity tokens with surrounding contextual tokens.

3) Effect of $\mathcal{L}_{\text{Fine}}$: Adding the fine-grained entity contrastive loss produces further mR improvements of 0.55, 0.65, and 0.70 percentage points on RSITMD, RSICD, and

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TABLE V
ABLATION STUDIES CONDUCTED WITH DIFFERENT COMPONENTS ON THREE DATASETS

Components	RSITMD				RSICD				UCM-Captions			
	Baseline	+C(V)+V	+C(T)+N	+$\mathcal{L}_{fine}$	Baseline	+C(V)+V	+C(T)+N	+$\mathcal{L}_{fine}$	Baseline	+C(V)+V	+C(T)+N	+$\mathcal{L}_{fine}$
CEBA(V)+VEBC		✓	✓	✓		✓	✓	✓		✓	✓	✓
CEBA(T)+NTIC			✓	✓			✓	✓			✓	✓
$\mathcal{L}_{fine}$				✓				✓				✓
Image R@1 (%)	20.22	23.50	24.73	**24.20**	13.84	14.22	14.63	**13.76**	14.48	18.95	23.62	**24.19**
Image R@5 (%)	53.27	55.49	56.90	**57.39**	37.53	39.41	40.36	**38.30**	60.10	65.24	71.14	**70.76**
Image R@10 (%)	71.19	71.46	74.82	**74.82**	52.34	55.44	54.74	**53.92**	93.81	91.71	97.24	**97.43**
Text R@1 (%)	26.55	26.99	27.88	**28.76**	15.26	14.95	16.87	**16.93**	16.10	25.24	25.71	**24.29**
Text R@5 (%)	45.13	44.69	51.55	**53.54**	25.38	25.18	33.04	**35.86**	54.67	59.52	57.62	**63.81**
Text R@10 (%)	58.63	58.41	65.27	**65.71**	36.52	36.83	44.49	**49.22**	80.86	78.57	83.81	**82.86**
mR (%)	45.83	46.76	50.19	**50.74**	30.14	31.01	34.02	**34.67**	53.33	56.54	59.86	**60.56**

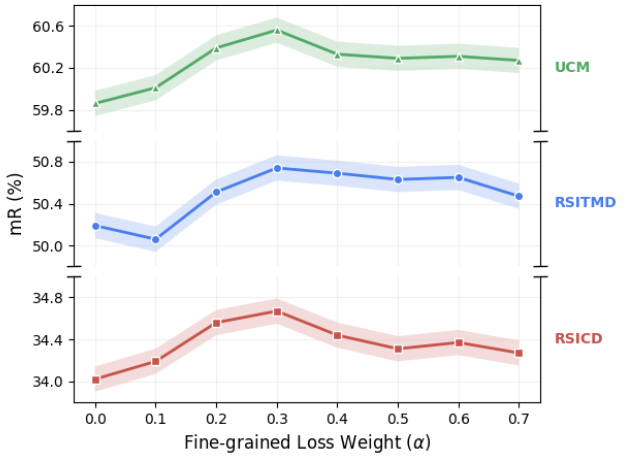


Fig. 6. Effect of the fine-grained loss weight α on mR across RSITMD, RSICD, and UCM-Captions. Shaded regions denote standard deviation over three independent runs.

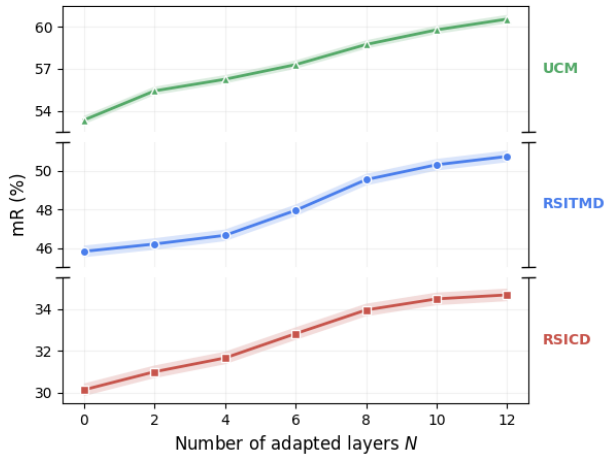


Fig. 7. Effect of the number of CEBA-adapted Transformer layers N on mR across RSITMD, RSICD, and UCM-Captions. Shaded regions denote standard deviation over three independent runs.

UCM-Captions, respectively. Without $\mathcal{L}_{\text{fine}}$, the entity representations are optimized only indirectly through the global contrastive objective. The additional gains show that sample-level entity contrastive supervision helps consolidate the cross-modal correspondences constructed by VEBC and NTIC.

4) Overall: From the baseline to the complete BEAN model, mR increases by 4.91, 4.53, and 7.23 percentage points on RSITMD, RSICD, and UCM-Captions, respectively. Each component provides an additional improvement when introduced in sequence, indicating that the visual entity path, textual entity path, and fine-grained supervision complement one another. These results support the effectiveness of combining VEBC, NTIC, CEBA, and $\mathcal{L}_{\text{fine}}$ for fine-grained image–text retrieval.

E. Hyperparameter Sensitivity Analysis

We investigate the sensitivity of BEAN to two hyperparameters: the weight of the fine-grained loss, α, and the number of Transformer layers equipped with CEBA, N. The results are shown in Figs. 6 and 7, respectively.

1) Fine-Grained Loss Weight α: We evaluate α over the set $\{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7\}$. When $\alpha = 0$, the fine-grained loss is removed, and the resulting mR values are 50.19%, 34.02%, and 59.86% on RSITMD, RSICD, and UCM-Captions, respectively. As α increases, the performance generally improves and reaches its best value at $\alpha = 0.3$, with mR values of 50.74%, 34.67%, and 60.56% on the three datasets. Further increasing α leads to a gradual performance decrease. For example, at $\alpha = 0.7$, the mR values decrease to 50.47%, 34.27%, and 60.27%, respectively. This result suggests that assigning excessive weight to $\mathcal{L}_{\text{fine}}$ may weaken the contribution of the global contrastive objective and affect scene-level alignment. The performance variation for $\alpha \in [0.3, 0.7]$ is relatively small, indicating that BEAN is not highly sensitive to moderate changes in the fine-grained loss weight. We therefore set $\alpha = 0.3$ in all subsequent experiments.

2) Number of Adapted Layers N: We evaluate the insertion of CEBA into the last N Transformer layers, where $N \in \{0, 2, 4, 6, 8, 10, 12\}$. When $N = 0$, CEBA is disabled, and the model reduces to the frozen CLIP baseline, yielding mR values of 45.83%, 30.14%, and 53.33% on RSITMD, RSICD, and UCM-Captions, respectively. Increasing the number of adapted layers consistently improves performance. Even when CEBA is inserted into only the deepest two layers ($N = 2$), the mR increases by 0.38, 0.86, and 2.09 percentage points on the three datasets. With $N = 8$, the corresponding mR values reach 49.54%, 33.95%, and 58.74%, which are close to the results obtained with CEBA in all 12 layers. The largest

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marginal improvements are observed when N increases from 6 to 10, while the gains become smaller for the shallowest layers. The best results are obtained at $N = 12$, suggesting that layerwise entity calibration remains beneficial throughout the Transformer encoder. Accordingly, we use the full-depth configuration, $N = 12$, as the default setting.

V. Conclusion

In this article, we present BEAN, a parameter-efficient framework for RS image–text retrieval designed to resolve the critical structural limitations of existing approaches: semantic ambiguity, alongside the severe dilution of entity signals within global embeddings and supervision deficits. To tackle the bottleneck of entity dilution, VEBC and NTIC are introduced to construct semantically unambiguous, cross-modal entity paths. This is coupled with CEBA to drive layerwise bidirectional enhancement between global and entity representations. In addition, the fine-grained entity contrastive loss ($\mathcal{L}_{fine}$) is proposed to provide direct sample-level supervision for explicit entity alignment without requiring manual positive-negative pair annotation. Extensive evaluations confirm that BEAN achieves leading performance among PEFT-based methods on the RSITMD, RSICD, and UCM-Captions benchmarks.

Despite its efficacy, the current framework relies on GroundingDINO and spaCy for offline entity preprocessing, which may limit its robustness against highly specialized RS terminology and densely cluttered scenes. Future work will explore end-to-end entity-aware adaptation to eliminate these offline dependencies, and extend the BEAN architecture to broader RS vision-language applications, including visual question answering and image captioning.

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